

# SpatioTemporal 3D Gaussian Splatting Alignment via Robust Features and Randomized Sampling

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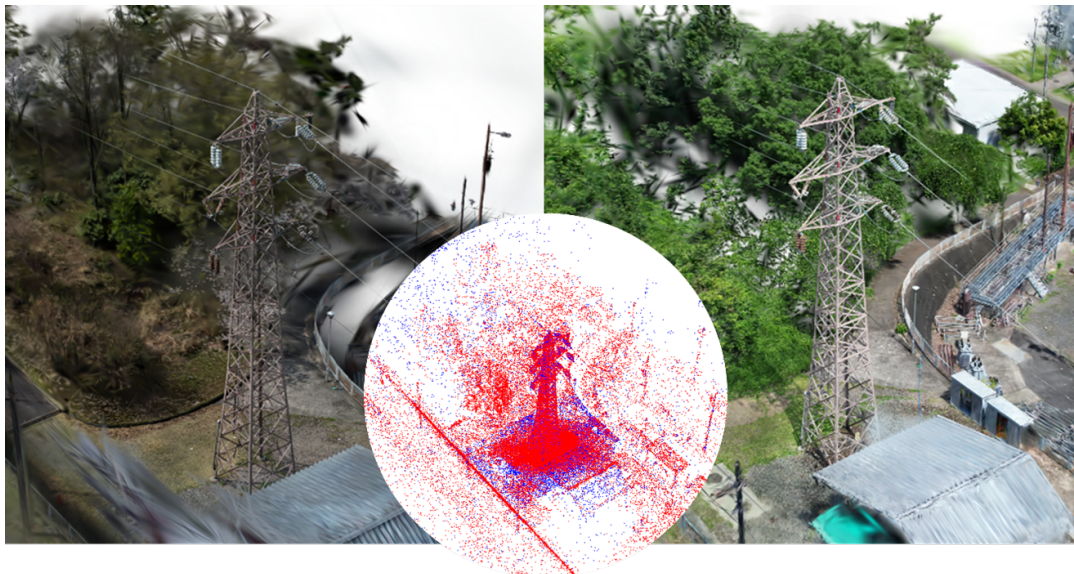


Figure 1: Alignment result around the tower in different seasons.

## Abstract

This study applies 3D Gaussian Splatting (3DGS) [Kerbl et al. 2023] to infrastructure maintenance, addressing the challenge of inconsistent coordinate systems in time-series data. While LiDAR is accurate, 3DGS offers quick, lightweight, photorealistic models via

smartphones. We propose a novel method leveraging the unique density of 3DGS points: sampling and cleaning efficiently extract key points. Applying TEASER++ [Yang et al. 2020] achieves fast, high-accuracy alignment even without a GPU. Experiments confirm our method’s superiority in speed and accuracy over existing techniques, enhancing 3DGS’s industrial practicality.

\*Equal contribution.

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## 1 Introduction

Japan's power infrastructure maintenance relies on 2D plans. 3DGS, using smartphones and so on, generates lightweight, photorealistic 3D models quickly. This allows efficient condition checks from any angle and monitoring of time-series changes. However, 3DGS creates models with inconsistent coordinate systems and scales, requiring manual alignment for chronological comparison. To improve practicality, this study proposes a method for fast and lightweight alignment of 3DGS data captured at different times.

## 2 Methodology

General methods fail to resolve the inconsistent scale and global position of individual 3DGS models, often requiring manual alignment for time-series data with significant changes.

*Randomized Sampling + Cleaning:* The point cloud from 3DGS is uniquely dense in areas rich with photometric information. We leverage this by using sampling and outlier cleaning to efficiently extract key points needed for alignment. This approach mitigates feature changes and substantially reduces computational costs. Outliers are removed based on distance from the centroid using the threshold  $D_{th} = \mu + k\sigma$ . This process effectively retains crucial information even at low sampling rates.

*Adapt aligning method:* The extracted point clouds are aligned using the proven FPFH + KNN + TEASER++ method. By selectively extracting essential points based on 3DGS characteristics, the approach significantly reduces computational costs and achieves both fast and highly accurate registration.

## 3 Experiments

We evaluate the performance of the proposed method by comparing it with the FPFH + KNN + TEASER++ method, a conventional ICP-based approach and PhotoReg [Yuan et al. 2024]. We used both a high-end PC with a GPU and a low-end PC without a GPU to assess processing time and alignment accuracy. To validate the effectiveness of our proposed method, we used the following experimental environments. Two types of Windows 11 computing environments were prepared: **The high-end PC** was equipped with an Intel Core i9-14900KF CPU, a 64GB of RAM and an NVIDIA GeForce RTX 4080 SUPER GPU. **The low-end PC** was equipped with an Intel Core i5-8400 CPU, a 16GB of RAM and no dedicated GPU. For the evaluation, we used a 3DGS data (a) containing multiple objects (a fox, a gun, and a paprika) and a second 3DGS data (b) containing a single object with relatively few features for alignment purposes (a toy whale), as shown in Figure 2.

## 4 Results and Future Work

Figure 3 confirms that a 5% sampling rate yields the highest Fitness, proving that sampling effectively mitigates feature changes and significantly reduces processing time on the high-end PC. Table 1 compares our method with existing approaches. The results confirm our proposed method's significant superiority in both Fitness and processing time. Similar results on the low-end PC demonstrate that the method delivers sufficient performance even with severe computational resource constraints (Data not shown).

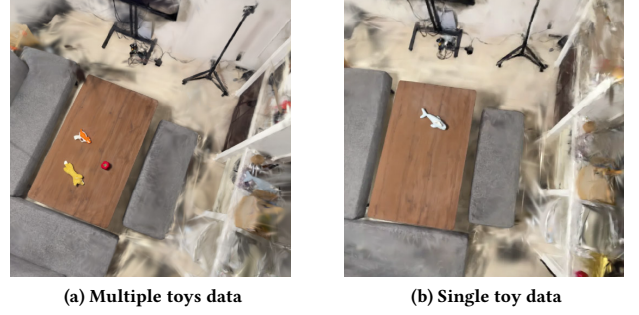


Figure 2: Datasets for experiments.

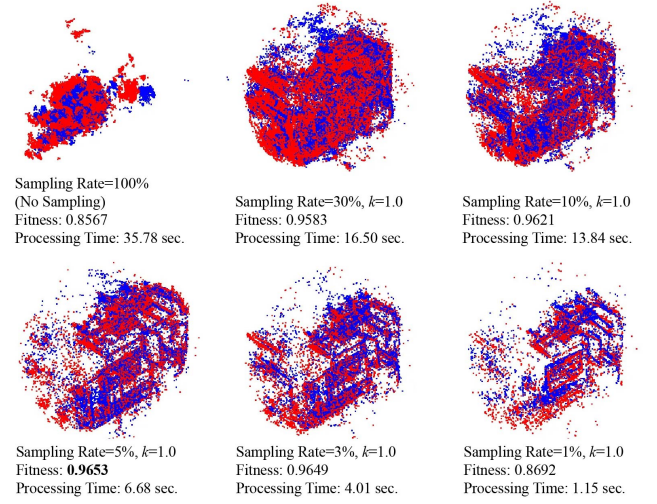


Figure 3: Results of alignment with different sampling rates.

Table 1: Frequency of Special Characters 2

| Method   | Fitness       | Processing Time(s) |
|----------|---------------|--------------------|
| Ours     | <b>0.9653</b> | <b>6.68</b>        |
| TEASER++ | 0.8567        | 35.78              |
| ICP      | 0.3198        | 94.16              |
| PhotoReg | 0.0408        | 1556.69            |

As a future work, We will apply this method to real world infrastructure in the future to validate its effectiveness.

## References

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